LIST OF TABLES

5.6 ECOSYSTEM MARKET MAP

FIGURE 22 ECOSYSTEM OF MICROBIOME SEQUENCING SERVICES MARKET

	SERVICE	TYPE	END USER
CORE SEGMENT	▪ Sample Preparation ▪ Sequencing and Library Preparation ▪ Data Analysis	▪ Amplicon Sequencing ▪ Shotgun Sequencing ▪ Nanopore Sequencing ▪ De Novo Sequencing and Others	▪ Pharmaceutical and Biotechnology Companies ▪ Academic and Research Institutes ▪ Other End Users

Source: MarketsandMarkets Analysis

5.7 PORTER'S FIVE FORCES ANALYSIS

TABLE 6 MICROBIOME SEQUENCING SERVICES MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	Low
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	Moderate
Threat of Substitutes	Low to moderate
Threat of New Entrants	Low to moderate

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.7.1 THREAT OF NEW ENTRANTS

The global microbiome sequencing services market is not a consolidated market. The well-established players operating in this market have narrow offerings, and a very small part of their portfolios are dedicated to microbiome sequencing. However, strong distribution channels and a global presence have made it easier for these players to maintain their market position. Moreover, existing product manufacturers have strong relationships with their customers (service providers) and are engaged in long-term contracts to supply the necessary equipment. As the industry demands high capital investments for R&D, testing, and manufacturing products, the entry barriers for new service providers in the market are estimated to be low to moderate.

5.7.2 THREAT OF SUBSTITUTES

The impact of this factor is low to moderate since there are fewer substitutes for established technologies in this market. The preference for traditional technologies for sequencing microbiomes may decrease due to the availability of advanced technology-oriented services.

5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers is moderate in this market. This market has witnessed significant technological innovations in developing affordable and user-friendly products. Thus, suppliers have better bargaining powers if their products are highly differentiated and endowed with breakthrough technologies. Products with a high degree of technological capabilities can command a premium price in the market. The high degree of technological capabilities, coupled with a high buyer-to-seller ratio, significantly increases the bargaining power of suppliers. Furthermore, compatibility issues between instruments and consumables (which are responsible for recurrent costs) increase the bargaining power of suppliers.

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers in the microbiome sequencing services market is moderate. Academic research institutes (one of the largest end users of sequencing products) are involved in bulk purchases of products and bulk sequencing of samples. This trend of bulk purchasing further increases the bargaining power of buyers in this market.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The microbiome sequencing services market is fragmented in nature, with the top 3 players accounting for a share of ~20%. A large number of local players and small-scale CROs account for the remaining market share. These players operate in niche market segments. Industry players are expected to cash in on the various growth opportunities and gain maximum share in the market. This has created low-entry barriers for new players, and the existing players compete to expand their shares.

5.8 REGULATORY ANALYSIS

5.8.1 NORTH AMERICA

5.8.1.1 US

NGS sequencing technologies are being used predominantly in microbiome and metagenome sequencing. The increase in use is further supported by increasing advancements in genomics. Several federal agencies regulate NGS products, including the Food and Drug Administration (FDA), the Centers for Medicare and Medicaid Services (CMS), and the Federal Trade Commission (FTC).

CMS Regulations:

The CMS regulates clinical laboratories, including laboratories conducting clinical genetic testing, through its Clinical Laboratory Improvement Amendments of 1988 (CLIA) program. The CLIA established a certification that laboratories must pass to conduct clinical testing legally. The objective of the CLIA is to determine clinical testing quality, including verification of the procedures used and the qualifications of the technicians processing the tests.

FDA Regulations:

Until recent years, the FDA has practiced enforcement discretion for laboratory-developed tests (LDTs), which implies that LDTs are used in the clinic without the FDA's assessment of their analytical and clinical validity. However, due to rapid advances in NGS, the growing need for clinical genetic testing, the growth of direct-to-consumer (DTC) genomic testing, and concerns that unregulated tests pose a public health threat, the FDA is modifying its approach.

The FDA has developed new guidance to regulate NGS genetic tests and verify their analytical and clinical validity. The agency has also drafted guidance proposing a new regulatory framework for LDTs. However, FDA guidance only represents the FDA's current thinking on a topic and is not legally binding for the FDA or the parties it regulates. However, adhering to FDA guidance is beneficial because it can streamline the regulatory process. The draft guidance is listed below. Since they are in draft form, they are not currently being implemented.

List of FDA Draft Guidance on Genetic Tests:

- Framework for Regulatory Oversight of Laboratory Developed Tests (2014)
- FDA Notification and Medical Device Reporting for Laboratory Developed Tests (2014)
- Discussion Paper on Laboratory Developed Tests (2017)
- Next-generation Sequencing (NGS) Draft Guidance